

Ashim Dhor

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About Me

I work at the intersection of interpretability and applied AI systems, spanning mechanistic interpretability of transformers, sequence models, and multi-agent systems. I am interested in understanding how these models behave, where they can fail, and how we can make them safer and more reliable. Going forward, I want to explore these questions in AI systems that can interact with the real world, particularly through computer vision and robotics. I would like to work towards building systems that are not only capable, but also safe and reliable.

Education

Indian Institute of Science Education and Research (IISER) Bhopal

Madhya Pradesh, India

BS-MS in Data Science and Engineering, CPI: 7.17/10

Dec 2021 - Present

Research Experience

Research Collaborator

Jan 2026 - Present

with Dr. Pin-Yu Chen, IBM MIT-IBM Watson AI Lab

- Co-developed a unified sequence architecture in which convolution, self-attention, and autoregressive recurrence appear as special cases of one learnable integral kernel; wrote custom Triton kernels for tiled fusion and Monte Carlo integration to make training tractable. **Outcome:** arxiv.org/abs/2606.19538.
- Proved that a transformer's Koopman spectrum (a spectral signature of how its layer-to-layer dynamics evolve) is an invariant of the model itself, not an artifact of parameterization, and that it can be recovered at the minimax-optimal statistical rate; validated on three pretrained transformers. **Outcome:** arxiv.org/abs/2608.10172.

AI Safety Research Scientist

August 2026 - Present

Lexsi Labs

- Working on multilingual LLM safety, building circuit-localised weight-editing methods (C- $\Delta\Theta$ [🔗](#)) and India grounded safety benchmarks to measure and repair harmful-compliance gaps across English and five Indic languages.
- Building activation-steering and circuit-editing pipelines for open-weight LLMs, using leakage safe, registry anchored contrast datasets to selectively correct hazardous chem/bio assistance & identity-based bias (caste, religion, region, etc).

AI Engineer Intern

Jan 2026 - July 2026

Elucidata Data Consulting Pvt. Ltd.

- Developed a production ready AI agentic system for flow cytometry analysis: end-to-end pipeline (FCS ingestion $\rightarrow$ QC gating $\rightarrow$ normalization $\rightarrow$ cell population identification) deployed across heterogeneous multi-sample datasets.
- Processed and integrated multi-omics datasets (transcriptomics, proteomics, flow cytometry) with structured clinical metadata - hands-on experience with EHR-adjacent data pipelines and cross-modal data alignment.

Master's Thesis Researcher

Jul 2025 - Present

Data Science Lab, IISER Bhopal (collab: AIIMS Bhopal, JNCHRC Bhopal)

- Designed and led a multi-agent oversight framework: used NMF on four foundation-model ensembles to discover interpretable latent concepts, trained concept-specialized ViT-Small agents on concept-pure vs. mixed curricula, and implemented query-aware routing with a Chain-of-Discussion protocol. **Outcome:** [Paper 1](#), [Paper 2](#), [Paper 3](#)

Research Intern - Quantum ML for Fraud Detection

Jun 2023 - Sept 2023

QWorld

- Architected a Quantum Graph Neural Network with Topological Data Analysis and Variational Quantum Circuits; achieved 94.5% accuracy and 0.85 PR-AUC, outperforming classical GraphSAGE (92.4%, 0.77). **Outcome:** [Paper](#)

Selected Publications

[Ashim Dhor](#), Rasel Mondal, Tanmay Basu. **MOSAIC: Morphologically Orchestrated Specialized Agents for Interpretable Cancer Diagnosis** ([link](#)). *IEEE ICHI 2026*, Young Scholar Symposium. [Extended abstract]

[Ashim Dhor](#), Abhirup Banerjee, Tanmay Basu. **TUNE++: Topology-Guided Uncertainty Estimation for Reliable 3D Medical Image Segmentation** ([link](#)). *MIDL 2026*. [Full Paper]

[Ashim Dhor](#), Emily Das, Rasel Mondal, et al. **HISTO-UNet: Histopathology Image Segmentation Using Topology-Aware UNet with Dual Uncertainty Quantification** ([link](#)). *IEEE ISBI 2026*. [Full Paper]

[Ashim Dhor](#), Smily Bharadwaj. **Topo-GraT: Learning to Grow with Causal Graph Transformers** ([link](#)). *AAAI 2026*. [Student Abstract Track]

Nouhaila Innan, Abhishek Sawaika, [Ashim Dhor](#), et al. (2024). **Financial fraud detection using quantum graph neural networks** ([link](#)). *Quantum Machine Intelligence*. [Full Paper]

Work Submitted

Identifiable but Not Legible: Spectral Structure in Mechanistic Interpretability[†]

Under review in AAAI

The identifiability theorem for a mechanistic-interpretability primitive: recasting the transformer forward pass as a controlled dynamical system and lifting it with the Koopman operator yields a spectrum that is a coordinate-free model invariant, recoverable at the minimax-optimal rate $M^{-1/2}$. Validated on GPT-2 small, Gemma-2-2B, and Qwen3-8B-Base.

Unified Neural Operator (UNO): Learnable Integral Transform for Sequence Modeling[†]

Under review in NeurIPS

A single learnable-kernel architecture, with a custom Triton kernel for the pairwise-interaction MLP, that recovers convolution, self-attention, and autoregressive recurrence as special cases. UNO matches or exceeds specialized baselines on ImageNet-1K, GLUE, ModelNet40, VQA v2, and NLVR2; architectural inductive bias produces qualitatively different OOD failure modes.

Collaborative Multi-Agent Reasoning for Whole-Slide Histopathology[†]

Under review in WACV

Extended version of MOSAIC paper. Non-negative matrix factorization over an ensemble of foundation models decomposes the representation space into six morphological concepts, which are then used to build concept-separated training curricula for a set of lightweight ViT-Small student agents. At inference, a query-aware router selects which agents are relevant to a given region, and a Chain-of-Discussion protocol runs multiple rounds of evidence exchange among the selected agents.

Diagnosing and Mitigating Cross-Attention Collapse in Multimodal VQA[†]

Under review in WACV

Diagnostic framework for language-prior bias in VQA: a per-branch, per-epoch metric quantifying when cross-attention becomes image-invariant, a no-parameter regularizer restoring image-dependent attention, and causal probes distinguishing grounding from regularization.

[†] *Titles simplified to preserve submission anonymity.*

Awards & Recognitions

NeurIPS 2026 - Selected in Affinity Support Pilot Program for NeurIPS 2026 (link)	2026
Google DeepMind APAC Research Symposium 2026 - Selected for the Student Symposium	2026
ASDP 2026 Travel Grant - 3rd Annual Congress of ASDP 2026, Singapore (1300 SGD) (link)	2026
Rising Star Award - IEEE ICHI 2026 Young Scholar Symposium (link)	2026
MICCAI Society Membership Grant (100 USD) (link)	2026
Student Scholarship Travel Grant - AAAI 2026 (1100 USD) (link)	2026
Winner - HistoAI , National Bio Entrepreneurship Competition (NBEC)	2025
QIntern : Best Project and Best Presentation Award (link)	2023

Technical Skills

Infrastructure & Distributed Training: training on NVIDIA H200/A100 clusters, Weights & Biases, Git, Linux

Programming Languages: Python, C, MATLAB, SQL, $\text{\LaTeX}$, Qiskit; PyTorch, Hugging Face Transformers & PEFT

Alignment & Interpretability: LLM post-training (SFT, RLHF, RL fine-tuning), chain-of-thought faithfulness, deceptive-alignment detection, sparse autoencoders, thought anchors, monitorability evaluation

Multi-Agent & Evaluation Systems: multi-agent system design, query-aware routing, agentic pipelines, foundation model adaptation across vision/language/graph modalities

Service

Program Chair: AAAI 2027 Student Abstract and Poster Program

Reviewer: NeurIPS 2026 Workshop on TAE, NeurIPS 2026 Global South Program, ICML 2026 (Trustworthy AI for Good Workshop), CHIL 2025, ML4H Symposium 2025, AAAI 2026

Mentor: AAAI 2026 Undergraduate Consortium Program

References

Dr. Abhirup Banerjee - Associate Professor, University of Oxford

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Dr. Pin-Yu Chen - Principal Research Scientist, IBM MIT-IBM Watson AI Lab

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Dr. Tanmay Basu - Associate Professor, IISER Bhopal

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